

Intelligent Decision Support System for Sri Lankan Tea Estate Management

R26-IT-118

Project Proposal Report

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Sri Lanka Institute of Information Technology

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Supervised By: Mrs. Uthpala Samarakoon

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DECLARATION

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I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Tea leaf diseases significantly affect the productivity and quality of Sri Lanka's tea industry. Traditional disease identification in tea estates relies mainly on manual visual inspection by supervisors, which is time-consuming, inconsistent, and dependent on expert availability. Delayed or inaccurate diagnosis often leads to improper pesticide usage, increased operational costs, and yield losses. Furthermore, most existing research focuses on disease classification alone, without integrating severity estimation, treatment guidance, or long-term pattern analysis.

This research proposes a Tea Leaf Disease Detection and Pattern Analysis System as a core component of an Integrated AI-Driven Intelligent Decision Support System for tea estate management. The system utilizes Convolutional Neural Networks (CNN) to automatically classify tea leaf diseases using images uploaded via a web application. A tea-specific disease image data set is created and labeled to train and evaluate the CNN model for disease type and severity prediction. Performance is assessed using standard evaluation metrics such as accuracy, precision, recall, and confusion matrix analysis.

Beyond real-time diagnosis, the system incorporates a treatment recommendation module that provides suggested interventions based on detected disease type and severity level. Additionally, historical disease records are analyzed to identify seasonal trends, location-based hot spots, and recurring outbreak patterns, enabling predictive disease management rather than isolated detection.

By combining CNN-based classification, severity estimation, treatment guidance, and pattern analysis, this research contributes a comprehensive and context-aware disease management solution tailored to Sri Lankan tea estates. The proposed approach enhances early intervention, reduces crop losses, and supports sustainable, data-driven estate management practices.

Keywords

Tea Leaf Disease Detection, CNN, Machine Learning, Pattern Analysis, Severity Prediction, Treatment Recommendation, Decision Support System

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LIST OF ABBREVIATIONS

Abbreviation	Full Meaning
AI	Artificial Intelligence
ML	Machine Learning
CNN	Convolutional Neural Network
DSR	Design Science Research
API	Application Programming Interface
UI	User Interface
SDG	Sustainable Development Goal
TIM	Technology Integration and Management
GPU	Graphics Processing Unit
SaaS	Software as a Service
VPS	Virtual Private Server

INTRODUCTION

The tea industry is one of the most significant contributors to Sri Lanka's national economy, providing employment opportunities, foreign exchange earnings, and supporting rural livelihoods. Maintaining high tea quality and consistent production is critical for sustaining global competitiveness. However, tea leaf diseases remain a major challenge affecting yield, leaf quality, and overall plantation productivity. Common diseases such as blister blight, red rust, and grey blight can spread rapidly under favorable environmental conditions, causing substantial crop losses if not detected and treated at an early stage.

In most Sri Lankan tea estates, disease detection primarily depends on manual visual inspection performed by field supervisors or estate managers. This traditional approach is time-consuming, subjective, and highly dependent on expert experience. Limited access to specialists and delays in identifying disease severity often result in improper treatment decisions, excessive pesticide usage, and increased operational costs. Furthermore, manual systems lack structured data recording mechanisms, making it difficult to analyze long-term disease patterns or predict future outbreaks.

Advancements in Artificial Intelligence (AI) and Machine Learning (ML), particularly Convolutional Neural Networks (CNN), have demonstrated strong capabilities in image-based classification tasks. In agricultural contexts, CNN models have been successfully applied to detect plant diseases with high accuracy. However, most existing research focuses solely on disease classification in controlled environments and generic crop data sets. There is limited work that integrates tea-specific disease detection with severity estimation, treatment recommendation, and historical pattern analysis within a unified decision support framework.

This research proposes a Tea Leaf Disease Detection and Pattern Analysis System as a component of an Integrated AI-Driven Intelligent Decision Support System for Sri Lankan tea estates. The system enables users to upload tea leaf images via a web-based application. A CNN-based model performs automated disease classification and severity prediction, while an additional analytical module evaluates historical disease data to identify seasonal trends and recurring outbreak patterns. A treatment recommendation mechanism further supports timely and appropriate intervention.

By combining real-time disease detection with predictive pattern analysis and decision support features, the proposed solution aims to enhance early diagnosis, reduce crop losses, optimize chemical usage, and improve overall estate management efficiency. This research contributes toward a sustainable and data-driven approach to tea disease management tailored specifically to the Sri Lankan tea industry.

1.1 Background and Context

Tea is one of Sri Lanka's most valuable agricultural exports, contributing significantly to national income, rural employment, and foreign exchange earnings. Sri Lanka produces approximately 300 million kilograms of tea annually, making it one of the world's leading tea exporters. Maintaining high leaf quality and stable yield is therefore essential for sustaining the country's global reputation in premium tea production. However, tea leaf diseases pose a major threat to plantation productivity and tea quality. Common diseases such as blister blight, red rust, and grey blight spread rapidly under favorable climatic conditions, particularly in humid and high-altitude regions. These diseases damage young shoots and mature leaves, reduce photosynthetic efficiency, and significantly affect both harvest yield and processed tea quality.

Currently, disease detection in many tea estates relies primarily on manual visual inspection by field supervisors. This traditional method is time-consuming, subjective, and highly dependent on the experience of individual inspectors. Early-stage symptoms may go unnoticed, leading to delayed intervention and increased disease spread. Large plantation areas also make continuous monitoring difficult. As a result, estate managers, field supervisors, plantation workers, and tea factory operators are directly affected by inefficient disease detection and delayed management actions.

Although recent research has explored Artificial Intelligence and deep learning techniques for plant disease detection, many existing solutions focus mainly on laboratory datasets and disease classification accuracy, without integrating severity prediction, treatment recommendation, or long-term disease pattern analysis in real plantation environments.

To address these challenges, this research proposes a CNN-based Tea Leaf Disease Detection and Pattern Analysis system that enables automated disease identification, severity estimation, and data-driven disease trend analysis to support proactive tea estate management.

This research aligns with the Sustainable Development Goals (SDG 9 – Industry, Innovation and Infrastructure, and SDG 12 – Responsible Consumption and Production) by promoting technology-driven agricultural management and sustainable plantation practices. The study also belongs to the Technology Integration and Management (TIM) research cluster, focusing on integrating Artificial Intelligence technologies into practical plantation management systems.

1.2 Literature Review

Artificial Intelligence and deep learning technologies have increasingly been applied in agriculture to improve plant disease detection, crop monitoring, and yield management. Image-based disease identification using Convolutional Neural Networks (CNNs) has become one of the most effective techniques for detecting plant diseases automatically. These technologies reduce reliance on manual inspection and improve the accuracy and speed of disease diagnosis.

Mohanty et al. (2016) demonstrated the effectiveness of deep learning for plant disease detection using CNN models trained on a large dataset of plant leaf images. The study achieved high classification accuracy across multiple crop diseases and proved that CNNs are suitable for automated disease recognition systems [1]. However, the dataset mainly consisted of controlled environment images and generic crop types rather than tea-specific plantation data.

Santhiya and Radhakrishnan (2025) proposed TEADIS, a deep learning-based system designed specifically for tea leaf disease classification and segmentation. Their research used field observation images to improve real-world applicability and demonstrated promising results in detecting tea leaf diseases [2]. Nevertheless, the system focused mainly on disease classification and did not incorporate severity estimation or treatment recommendation mechanisms.

Zhang et al. (2023) explored the use of deep convolutional neural networks and transfer learning for automated tea quality identification. Their research showed that CNN models can effectively extract visual features such as color, texture, and shape from tea leaves to perform classification tasks [3]. Although this work demonstrated the strength of deep learning in tea leaf analysis, the study mainly focused on tea grading rather than disease detection.

Blasco et al. (2003) developed a machine vision system for automatic quality grading of agricultural products using traditional image processing techniques. The system used feature extraction methods to classify agricultural products based on visual characteristics [4]. While this research demonstrated the early potential of automated visual inspection systems, it relied on handcrafted features rather than modern deep learning approaches.

Kamilaris and Prenafeta-Boldú (2018) conducted a comprehensive survey of deep learning applications in agriculture. Their review highlighted that CNN-based models are widely used for crop monitoring, disease detection, and agricultural image analysis [5]. However, many systems remain experimental and lack integration with real agricultural management systems.

Batool et al. (2022) investigated machine learning models for predicting tea crop yield using climate and environmental data. Their work demonstrated the potential of data-driven models for plantation management and agricultural decision support [6]. However, the study focused on yield prediction rather than visual disease detection.

Mampitiya et al. (2025) proposed an explainable artificial intelligence approach for estimating Sri Lankan tea yield. Their research emphasized the importance of AI-based analytics in improving plantation management and agricultural forecasting [7]. However, the study did not address image-based disease detection or early disease monitoring.

In addition to academic research, several commercial agricultural technology platforms provide digital crop monitoring services. Applications such as Plantix and CropIn allow farmers to upload plant images and receive disease diagnosis and crop management advice. Similarly, agricultural data platforms such as AgriWebb provide digital tools for farm management and data tracking. However, these systems are generally designed for generic crop environments and do not provide specialized solutions tailored to tea plantation disease detection or integrated plantation decision-support systems.

Overall, existing studies demonstrate the strong potential of deep learning and artificial intelligence in agricultural monitoring and plant disease detection. However, most existing approaches focus primarily on classification accuracy or single-domain applications and lack integrated solutions designed specifically for tea plantation management.

1.3 Research Gap

Although recent studies have demonstrated the effectiveness of deep learning techniques such as Convolutional Neural Networks (CNNs) for plant disease detection, most existing approaches focus primarily on image-based classification tasks. For example, several studies successfully identify plant diseases from leaf images with high accuracy, but their systems are typically limited to disease recognition only. These solutions often rely on controlled datasets and do not consider the broader decision-making needs of plantation management.

Furthermore, existing research rarely integrates additional analytical capabilities such as disease severity estimation, treatment recommendation mechanisms, or long-term disease pattern analysis. As a result, current systems provide limited support for proactive disease management in real plantation environments. Estate managers and supervisors still depend heavily on manual expertise to determine treatment strategies and to monitor recurring disease trends.

Therefore, there is a need for a more comprehensive AI-driven solution that goes beyond simple disease identification. The proposed research addresses this gap by developing a tea-specific CNN-based disease detection system that integrates disease classification, severity estimation, treatment recommendation, and historical pattern analysis to support intelligent and proactive tea plantation disease management.

Table 1.1 presents a comparison of the existing research methods and the proposed system, highlighting the unique features and contributions of this research relative to prior work.

As shown in Table 1.1, the proposed system is the only approach that integrates CNN-based tea-specific disease detection with severity prediction, treatment recommendation, disease pattern analysis, and real-time web-based implementation, addressing the key limitations identified in existing research.

Table 1.1 Comparison of the existing research methods and proposed method

Technologies and Methods	[1] Mohanty et al. (2016)	[2] Santhiya & Radhakrishnan (2025)	[3] Zhang et al. (2023)	[4] Blasco et al. (2003)	[5] Mampitiya et al. (2025)	Proposed Method
CNN-based image classification	✓	✓	✓	✗	✗	✓
Tea-specific disease detection	✗	✓	✗	✗	✗	✓
Real plantation image dataset	✗	✓	✗	✗	✗	✓
Disease severity prediction	✗	✗	✗	✗	✗	✓
Treatment recommendation system	✗	✗	✗	✗	✗	✓
Disease pattern / trend analysis	✗	✗	✗	✗	✗	✓
Integration with plantation decision support	✗	✗	✗	✗	✓	✓
Web / real-time system implementation	✗	✗	✗	✗	✗	✓

2. OBJECTIVE

2.1 Main Objective

To design and evaluate a CNN-based intelligent system for detecting tea leaf diseases and analyzing disease patterns to support data-driven decision making in Sri Lankan tea plantations.

2.2 Specific Objectives

1. **To collect and construct a labeled tea leaf disease image dataset** representing common tea leaf diseases such as blister blight, red rust, and grey blight for training and evaluating deep learning models.
2. **To design and implement a Convolutional Neural Network (CNN) model** capable of accurately detecting and classifying tea leaf diseases from uploaded leaf images.
3. **To develop a disease severity estimation mechanism** that analyzes visual symptoms in leaf images to determine the level of infection and support early disease identification.
4. **To implement a treatment recommendation module** that suggests appropriate disease management actions based on the detected disease type and severity level.
5. **To integrate the disease detection system into a web-based application with a user-friendly interface**, enabling workers or supervisors to upload tea leaf images and obtain real-time disease detection results, severity levels, and treatment recommendations.
6. **To analyze historical disease detection data to identify seasonal and location-based disease patterns**, supporting data-driven decision making for proactive tea plantation disease management.

3.METHODOLOGY

3.1 Research Approach

This research follows the Design Science Research (DSR) methodology, which focuses on designing, developing, and evaluating an innovative technological solution to address a real-world problem. In this study, the artifact developed is Tea Leaf Disease Detection and Pattern Analysis system that supports disease identification and plantation decision-making. The research process includes problem identification, system design, model development, implementation, and evaluation.

The proposed system utilizes deep learning techniques to analyze tea leaf images and automatically identify disease types. The system is designed as a web-based decision support tool that allows plantation workers or supervisors to upload images of tea leaves and receive disease detection results. The system also estimates disease severity levels and provides treatment recommendations. In addition, historical disease records are analyzed to identify seasonal or location-based disease patterns that may assist plantation managers in proactive disease management.

3.2 Technical Methods and Algorithms

The core technical method used in this research is deep learning-based image classification using Convolutional Neural Networks (CNNs). CNN models are highly effective for analyzing image data because they automatically learn visual features such as texture, color, and shape from images.

This study applies transfer learning techniques using a pre-trained CNN model such as MobileNet or ResNet, which have been previously trained on large image datasets. Transfer learning allows the model to reuse learned features and adapt them to tea leaf disease classification tasks. The model is trained using labeled tea leaf images representing common diseases such as blister blight, red rust, and grey blight.

The system workflow includes image preprocessing, feature extraction using CNN layers, disease classification, and severity estimation. After detecting the disease type, a rule-based module provides appropriate treatment recommendations. The system also stores detection results in a database to support long-term disease pattern analysis.

3.3 Tools and Platforms

Several software tools and platforms are used to implement the proposed system.

The deep learning model is developed using Python, with machine learning libraries such as TensorFlow or PyTorch for model training and evaluation. Image preprocessing and feature extraction are supported using OpenCV.

The user interface is implemented using React, while Node.js is used for backend services and API integration. The system operates as a web-based application that can also be accessed through mobile devices, allowing field workers to upload leaf images directly from smartphones. Data storage and model outputs are managed through a cloud or database system that stores disease detection results and historical records.

These technologies were selected due to their strong support for AI development, scalability, and compatibility with modern web applications.

3.4 Data Collection and Ethical Considerations

The research utilizes both primary and secondary datasets. Primary data will be collected from tea plantations through manually captured images of tea leaves using mobile devices or cameras. These images will represent different disease conditions and healthy leaves.

Secondary datasets may also be used from publicly available agricultural image repositories such as Kaggle or research datasets used in previous studies. All collected images will be labeled according to disease type with guidance from domain knowledge and literature references.

Ethical considerations are taken into account during data collection. Images collected from plantations will not include personal or sensitive information. The collected data will be used solely for research and model training purposes.

3.5 Validation Strategy and Performance Metrics

The proposed CNN model will be evaluated using standard machine learning performance metrics. The dataset will be divided into training, validation, and testing sets to ensure reliable model evaluation.

The primary evaluation metrics include:

Accuracy – measures the overall correctness of disease classification

Precision – measures how many predicted disease cases are correctly identified

Recall – measures the ability of the model to detect actual disease cases

F1-score – provides a balanced evaluation of precision and recall

A confusion matrix will also be used to analyze classification performance across different disease categories. The performance of the proposed model will be compared with baseline classification methods to evaluate its effectiveness in detecting tea leaf diseases.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overview of the Proposed System

The proposed system is an Intelligent Decision Support System for Sri Lankan Tea Estate Management that integrates multiple intelligent components to support data-driven decision-making in tea plantations. The overall system consists of four main components: Labor Management and Welfare Optimization, Tea-Specific Climate and Weather Intelligence, Tea Leaf Disease Detection and Pattern Analysis, and AI-Based Tea Sorting and Grading.

The focus of this research is the Tea Leaf Disease Detection and Pattern Analysis component, which is responsible for identifying tea leaf diseases using image-based analysis. The system allows workers or supervisors to upload tea leaf images through a web-based interface accessible via desktop or mobile devices. The uploaded image is processed using a Convolutional Neural Network (CNN) model that detects disease type and estimates severity levels.

Once a disease is detected, the system generates treatment recommendations and stores the results in a database. The stored historical records are then used to analyze disease patterns across seasons or locations, enabling proactive plantation disease management.

4.2 System Components

The high-level system architecture consists of the following components:

1. **User Interface Layer**
A web-based interface that allows plantation workers or supervisors to upload tea leaf images and view disease detection results, severity levels, and recommended treatments.
2. **Image Processing and CNN Detection Module**
This module processes uploaded images using preprocessing techniques and applies a **CNN-based deep learning model** to detect tea leaf diseases.
3. **Disease Severity Estimation Module**
After disease classification, this module evaluates the severity level of the infection based on visual symptoms detected in the leaf image.
4. **Treatment Recommendation Module**
This component suggests suitable disease management actions based on the detected disease type and severity level.
5. **Disease Pattern Analysis Module**
Historical disease detection data is analyzed to identify **seasonal trends and recurring disease patterns** in tea plantations.

6. Database and Storage Layer

Stores uploaded images, detection results, severity information, and historical disease records for further analysis.

4.3 Component Interaction

The system components interact through a structured workflow. The user uploads a tea leaf image through the web interface, which is sent to the backend processing system. The image is first preprocessed and then passed to the CNN-based disease detection model.

The model predicts the disease type, after which the severity estimation module determines the level of infection. Based on this information, the treatment recommendation module generates appropriate management suggestions. The final results are returned to the user interface for display.

Simultaneously, the results are stored in the database for historical tracking and disease pattern analysis.

4.4 Data Flow Explanation

1. The user uploads a tea leaf image through the web interface.
2. The image is sent to the backend server for preprocessing.
3. The preprocessed image is passed to the CNN model for disease detection.
4. The system predicts the disease type and severity level.
5. The treatment recommendation module generates appropriate disease management suggestions.
6. The results are displayed to the user through the interface.
7. All detection results are stored in the database for pattern analysis and future reference.

4.5 Scalability and Security Considerations

The proposed system is designed to be scalable and capable of handling large volumes of image data from multiple tea estates. The use of cloud-based storage and modular system architecture allows the system to expand easily as more users and datasets are added.

Security measures are also considered to ensure safe system operation. Uploaded images and stored data are managed securely, and user access can be controlled through authentication mechanisms. Additionally, sensitive plantation data will be protected to maintain privacy and prevent unauthorized access.

The overall architecture of the proposed system is illustrated in Figure 4.1, showing the high-level structure of all major components and their interactions.

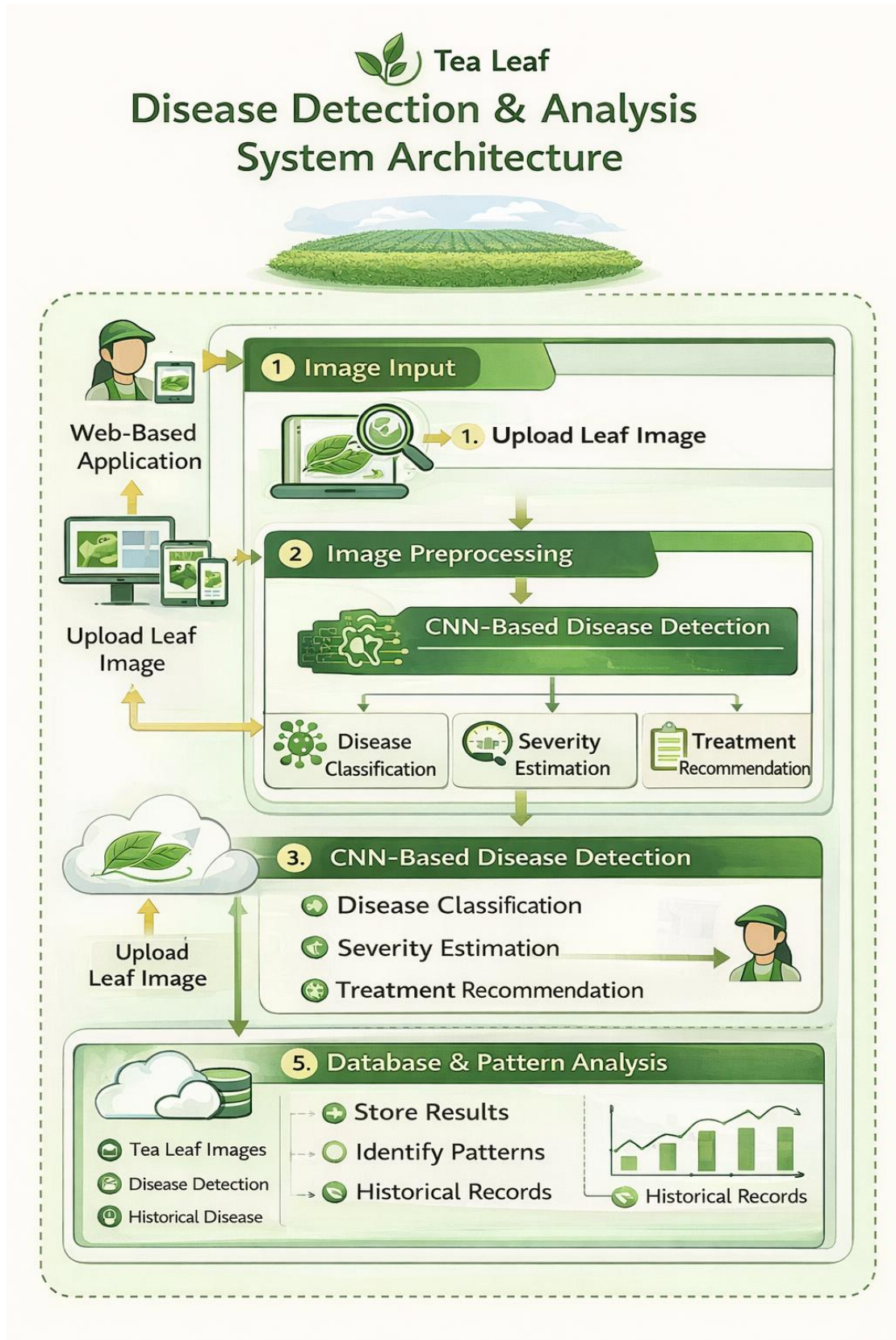


Figure 4.1 System Architecture Diagram

As illustrated in Figure 4.1, the system architecture follows a top-down flow beginning at the user interface layer where plantation workers or supervisors upload tea leaf images through the web-based application. The uploaded image passes through the image preprocessing stage,

which prepares it for CNN-based analysis by applying resizing, normalization, and noise reduction. The CNN disease detection module then processes the image to simultaneously produce three outputs: disease classification, severity estimation, and treatment recommendations, all of which are returned to the user in real time. At the base of the architecture sits the database and pattern analysis layer, which stores all detection records, historical disease images, and metadata. This layer continuously aggregates results to support the disease trend identification and location-based hotspot analysis that enable predictive plantation management.

5. USER REQUIREMENTS

3.6 Stakeholder Identification

The proposed Tea Leaf Disease Detection and Pattern Analysis system involves several stakeholders who interact with the system or benefit from its outputs.

Estate Managers are responsible for monitoring plantation health and making strategic decisions related to disease control, productivity, and resource allocation. They use the system to view reports, disease trends, and treatment recommendations.

Field Supervisors are responsible for inspecting tea plants in the field and identifying potential disease symptoms. They interact with the system by uploading tea leaf images and validating disease observations.

Tea Estate Workers support data collection activities by capturing images of infected leaves using mobile devices and submitting them through the application.

Researchers and System Developers use the system to analyze disease datasets, train machine learning models, and improve the accuracy of the CNN-based detection system.

These stakeholders collectively ensure the effective use of the system for accurate disease monitoring and plantation management.

3.7 Functional Requirements

FR1: The system shall allow users to upload tea leaf images through the web or mobile interface.

FR2: The system shall preprocess uploaded images to improve image quality and prepare them for CNN-based analysis.

FR3: The system shall classify tea leaf diseases using a trained Convolutional Neural Network (CNN) model.

FR4: The system shall estimate the severity level of detected diseases based on image features.

FR5: The system shall generate treatment recommendations based on the identified disease type and severity.

FR6: The system shall store uploaded images, detection results, and related metadata in a database.

FR7: The system shall analyze historical disease records to identify seasonal and location-based disease patterns.

FR8: The system shall provide dashboards and reports for estate managers to monitor disease trends and plantation health.

3.8 Non-Functional Requirement

Non-functional requirements describe system quality attributes such as performance, security, and usability.

Security

The system must ensure secure storage of uploaded images and user data. Access to administrative dashboards should be restricted through authentication mechanisms.

Performance

The disease detection model should provide prediction results within a few seconds after image upload to support timely decision-making in plantation environments.

Usability

The user interface must be simple and intuitive so that field supervisors and workers can easily upload images and view results without requiring advanced technical knowledge.

Reliability

The system should maintain consistent performance and accurate predictions when processing multiple image inputs.

Scalability

The system architecture should support future expansion to handle larger datasets and additional plantation locations.

6. COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

The primary target market for the proposed system includes tea estate management organizations, plantation companies, and agricultural research institutions in Sri Lanka. Large tea estates such as regional plantation companies require efficient monitoring tools to detect diseases early and reduce productivity losses.

The main customer persona includes estate managers and plantation supervisors responsible for maintaining tea plant health and ensuring consistent yield quality. These users require reliable and easy-to-use digital tools that support disease monitoring, early detection, and treatment decision-making.

Secondary users include agricultural researchers, government agencies, and tea industry regulatory bodies, which may use the system to analyze disease patterns and plantation health data at a regional level

6.2 Value Proposition

The proposed system provides a cost-effective AI-driven solution for early detection of tea leaf diseases using image analysis. Unlike traditional manual inspection methods, the system offers fast, accurate, and scalable disease identification.

Key value propositions include:

- Early detection of tea leaf diseases to reduce crop damage
- Automated disease classification using CNN-based AI models
- Severity estimation and treatment recommendations
- Data-driven disease pattern analysis for long-term plantation management
- Easy-to-use web/mobile interface for field supervisors

6.3 Revenue Model

The proposed system can adopt a Software-as-a-Service (SaaS) model, where tea estates subscribe to the platform on a monthly or yearly basis.

Possible revenue streams include

- Subscription plans for tea estates and plantation companies
- Premium analytics features for large-scale plantations
- Data insights services for agricultural research institutions
- Licensing the AI model to agricultural technology providers

6.4 Cost Estimation

The main costs involved in developing and deploying the system include:

- System Development Costs – AI model training, software development, and testing
- Infrastructure Costs – cloud storage, database hosting, and server processing
- Data Collection Costs – collecting tea leaf images and labeling disease datasets
- Maintenance Costs – system updates, security improvements, and model retraining

6.5 Pricing Strategy

A tiered pricing strategy can be implemented based on the size of the plantation and system usage.

Example pricing structure:

- **Basic Plan** – Small tea estates with limited image uploads and basic disease detection features
- **Standard Plan** – Medium estates with additional analytics and reporting tools
- **Enterprise Plan** – Large plantations with unlimited usage, advanced disease pattern analysis, and custom dashboards

6.6 Competitive Advantage

The proposed system offers several advantages compared to existing solutions:

- Tea-specific disease detection tailored to Sri Lankan plantations
- Integration of disease identification, severity estimation, and treatment recommendation in one system
- Ability to analyze historical disease patterns for predictive management
- Cost-effective solution compared to expensive industrial monitoring systems

6.7 Intellectual Property Considerations

The system may involve several intellectual property components, including:

- Proprietary CNN-based disease detection models trained on tea leaf datasets
- Custom-developed disease pattern analysis algorithms
- The integrated decision-support software platform

7. BUDGET AND JUSTIFICATION

Table 7.1 presents the estimated budget for the proposed system, including descriptions and total amounts in Sri Lankan Rupees for each major cost category.

As shown in Table 7.1, the total estimated budget for the proposed system development and deployment is Rs. 41,000, with cloud infrastructure and development tools accounting for the largest portion of the costs.

Table 7. 1 Estimated budget table

Description	Total Amount (Rs.)
Cloud Storage & GPU Hosting for CNN Model Training (Google Cloud / AWS – 6 months)	12,000
Image Dataset Storage & Backup Services	4,500
Web Application Hosting (Node.js / Flask Server – VPS)	6,000
Database Hosting (MongoDB / PostgreSQL Cloud Service)	3,000
AI Development Environment (Python libraries, model training tools, GPU runtime support)	3,500
Internet & Utilities for Development (shared cost for component)	3,000
Transport for Field Data Collection (tea leaf image capture from estates)	4,500
Dataset Annotation & Image Preparation Tools	2,500
Miscellaneous (unexpected costs, adapters, storage devices)	2,000
Total Budget	41,000

8. WORK BREAKDOWN STRUCTURE (WBS)

The table below outlines the key tasks of the Tea Leaf Disease Detection and Recommendation System project. It presents the task ID, description, timeline, and expected deliverables. The tasks progress from literature review and dataset preparation to CNN model development, system integration, testing, and final documentation, ensuring a structured and systematic project workflow.

Table 8. 1 Work Breakdown Structure

Task ID	Task Description	Timeline	Deliverable
T1	Conduct literature review on tea leaf disease detection using CNN and deep learning techniques	March	Literature review report
T2	Collect tea leaf disease image dataset from primary and secondary sources	March April	Tea leaf disease image dataset
T3	Label and preprocess images (resize, normalization, augmentation)	April	Preprocessed training dataset
T4	Design CNN architecture for tea leaf disease classification	May	CNN model architecture
T5	Train CNN model using labeled dataset	May – June	Trained disease detection model
T6	Evaluate model performance using accuracy, precision, recall, and confusion matrix	June	Model evaluation results
T7	Develop disease severity estimation and treatment recommendation module	July	Recommendation module
T8	Analyze historical disease data to identify seasonal and location-based patterns	August	Disease pattern analysis report
T9	Integrate CNN model with web application interface	September	Integrated system prototype
T10	Perform system testing and validation using new tea leaf images	October	Testing and validation report
T11	Prepare final documentation and proposal presentation	November	Final research report

The Work Breakdown Structure (WBS) diagram below presents the hierarchical organization of the Tea Leaf Disease Detection and Recommendation System project. It divides the project into four major pillars: Research and Data, Model Development, Post-Classification Features, and System Integration. Each pillar branches into specific sub-modules and tasks (T1–T11) to clearly define project activities and their logical workflow

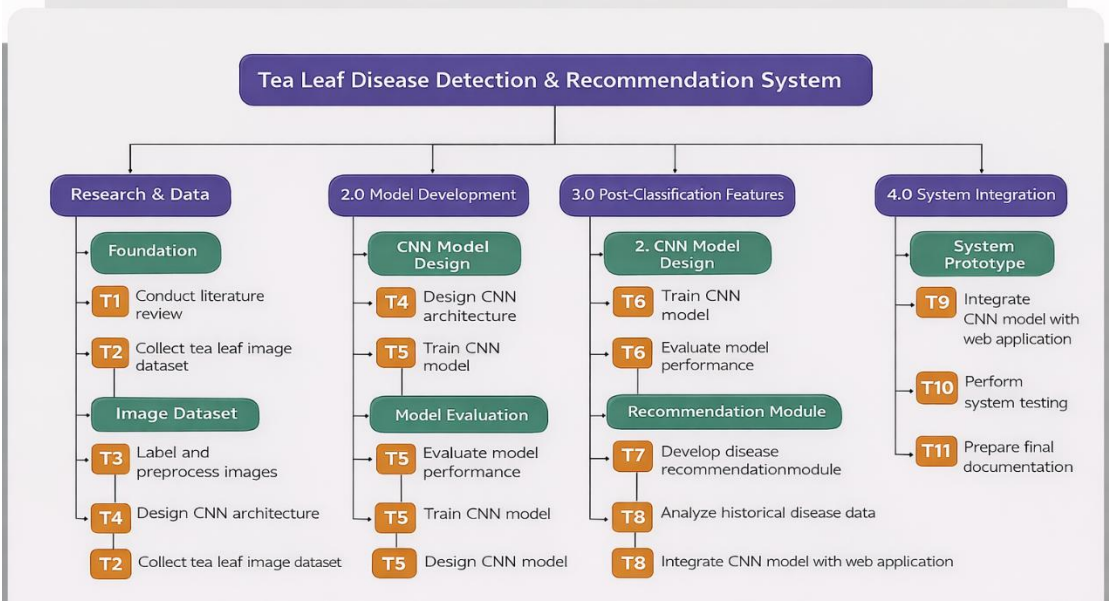


Figure 8. 1 Work Breakdown Structure

As shown in Figure 8.1, the WBS reveals the dependency relationships between tasks. The Research and Data pillar (T1–T3) must be completed first, as subsequent model development activities depend on the availability of a labeled, preprocessed dataset. The Model Development pillar (T4–T6) encompasses CNN architecture design, training, and evaluation and forms the technical core of the project. Post-Classification Features (T7–T8) extend the model outputs into actionable insights through severity estimation, treatment recommendation, and historical pattern analysis. Finally, System Integration (T9–T11) wraps all components into the web application, conducts end-to-end testing, and produces the final research documentation.

9.GANTT CHART

The project timeline is illustrated in Figure 9.1, which presents the Gantt chart showing the schedule for all eleven research tasks across the project period from March to November 2026.

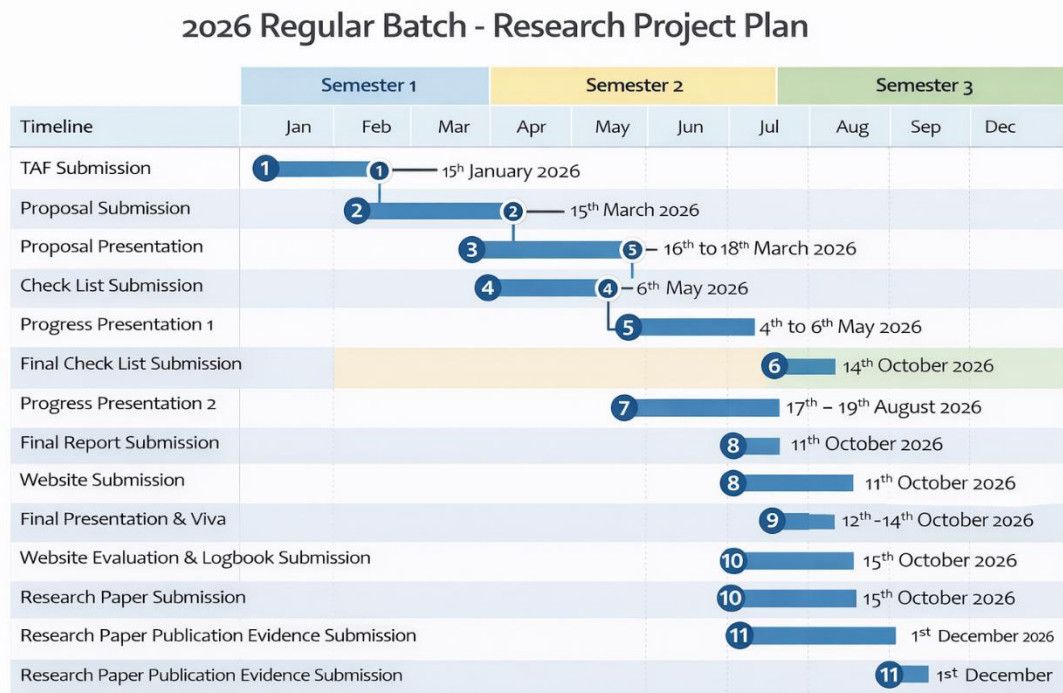


Figure 9.1 Gantt Chart

As illustrated in Figure 9.1, the project is organized across three academic semesters from January to December 2026, with clearly defined milestones at each phase. Semester 1 covers the foundational activities: the TAF Submission in January, Proposal Submission by 15th March, and Proposal Presentation from 16th to 18th March. These front-loaded activities correspond to the data collection, labeling, and preprocessing tasks (T1–T3). Semester 2 represents the technically intensive phase, encompassing CNN model design, training, and evaluation (T4–T6), with the Progress Presentation scheduled for 4th to 6th May 2026 serving as a key checkpoint. The severity estimation and treatment recommendation modules (T7) and historical pattern analysis (T8) also fall within this period, with the Check List Submission due by 6th May. Semester 3 consolidates all outputs into the integrated web prototype (T9), followed by system testing (T10) in October and final documentation (T11) in November. Critical submission deadlines in this phase include the Final Report Submission and Website Submission by 11th October, the Final Presentation and Viva from 12th to 14th October, and Research Paper Publication Evidence Submission by 1st December 2026.

10. REFERENCES LIST

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11. APPENDICES

11.1 Survey Instrument

Section A – Respondent Information

1. What is your role in the tea estate?
Estate Manager
Field Supervisor
Plantation Worker
Factory Operator
2. How many years of experience do you have in tea plantation management?
Less than 5 years
5–10 years
More than 10 years

Section B – Current Disease Monitoring Practices

3. How are tea leaf diseases currently identified in your estate?
Manual visual inspection
Laboratory testing
Expert consultation
Other: _____
4. How frequently are disease inspections conducted?
Daily
Weekly
Monthly
Only when symptoms appear
5. What are the most common tea leaf diseases observed in your estate?
Blister Blight
Red Rust
Grey Blight
Other: _____

Section C – Challenges in Disease Detection

6. What are the main challenges faced when identifying tea leaf diseases?
Lack of expert knowledge
Large plantation area
Delayed disease identification
Inconsistent diagnosis

7. How often do diseases spread before they are detected?
- Rarely
 - Sometimes
 - Frequently

Section D – Technology Adoption

8. Would an AI-based system that detects tea leaf diseases from images be useful for your estate?
- Yes
 - No
 - Not sure
9. Would you be willing to use a mobile or web-based application for disease detection?
- Yes
 - No
10. What additional features would you expect from such a system?
- Disease identification
 - Severity estimation
 - Treatment recommendations
 - Disease trend analysis

11.2 Detailed diagrams

Appendix Figure 11.2.1 presents the detailed system architecture diagram, showing all major modules and their interconnections within the proposed Tea Leaf Disease Detection and Pattern Analysis system.

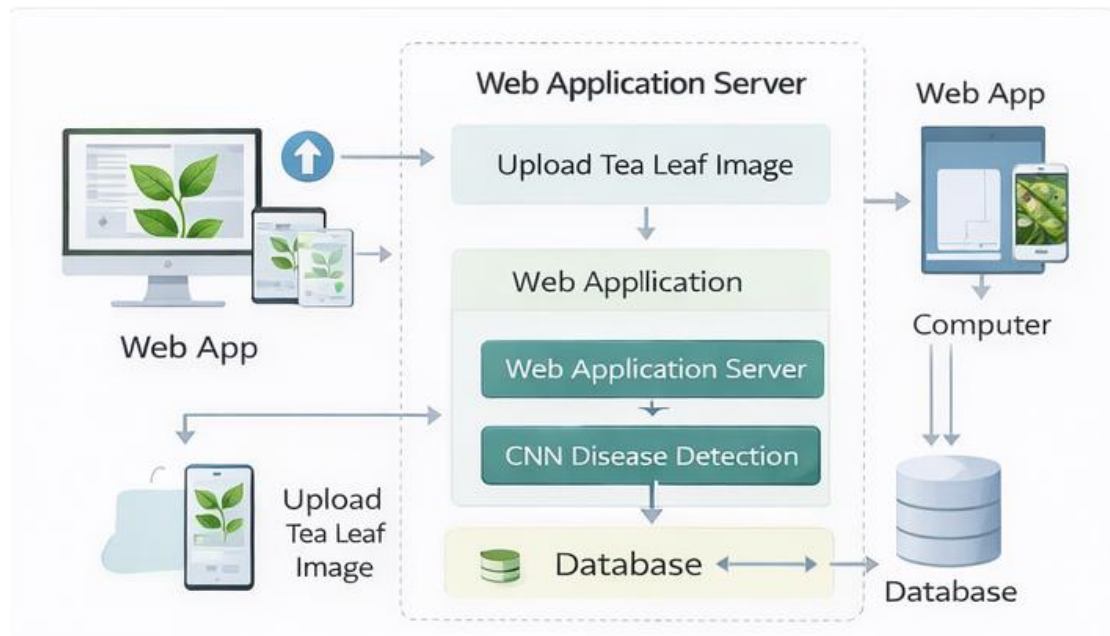


Figure 11.2.1– System Architecture Diagram

As shown in Figure 11.2.1, the architecture explicitly depicts the Web Application Server as the intermediary layer between the user-facing interface and the CNN disease detection engine. User devices (desktop and mobile) submit tea leaf images to the web application, which forwards them to the server for processing. The CNN module performs disease detection and returns results to the interface, while the database at the backend stores all images, detection outputs, and historical records for subsequent analysis.

Appendix Figure 11.2.2 illustrates the CNN disease detection workflow, tracing the step-by-step process from image input through preprocessing, feature extraction, classification, and output generation.

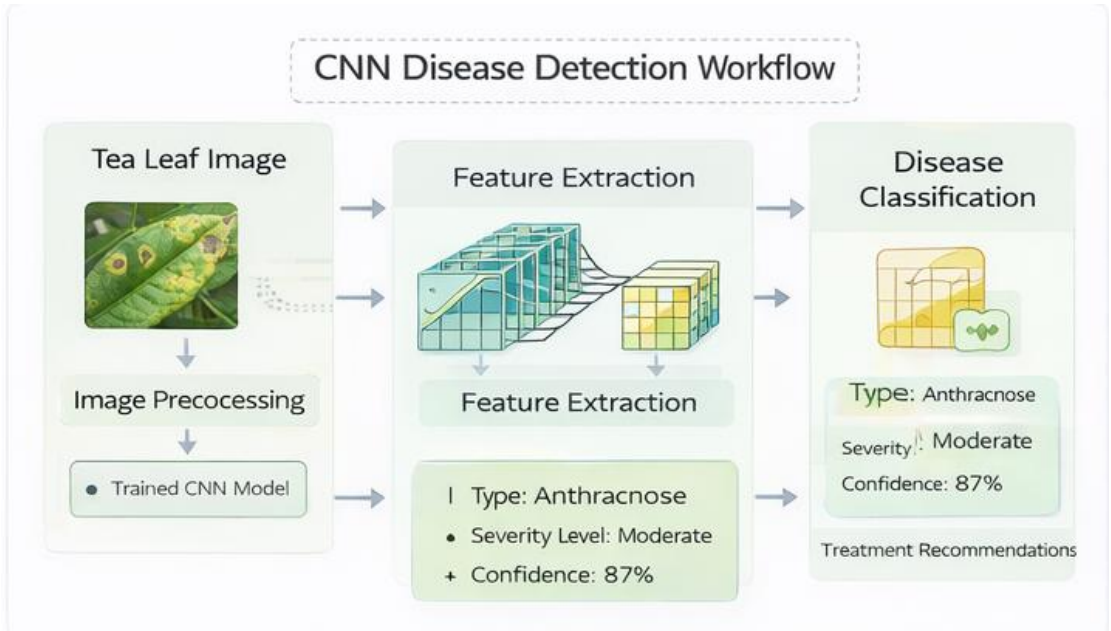


Figure 11.2.2– CNN Disease Detection Workflow

Figure 11.2.2 traces a single prediction cycle through the system. A raw tea leaf image is first submitted to the image preprocessing stage, where operations such as resizing, normalization, and contrast adjustment are applied to standardize the input. The preprocessed image is then passed through the trained CNN model, which extracts hierarchical visual features including color distribution, texture patterns, and lesion shape. The model outputs the detected disease type (e.g., Anthracnose), severity level (e.g., Moderate), and a confidence score (e.g., 87%), which collectively feed into the treatment recommendation module to generate targeted disease management advice.

Appendix Figure 11.2.3 presents the data flow diagram, showing how data moves through the system from the point of image upload to the generation of disease detection results and treatment recommendations.

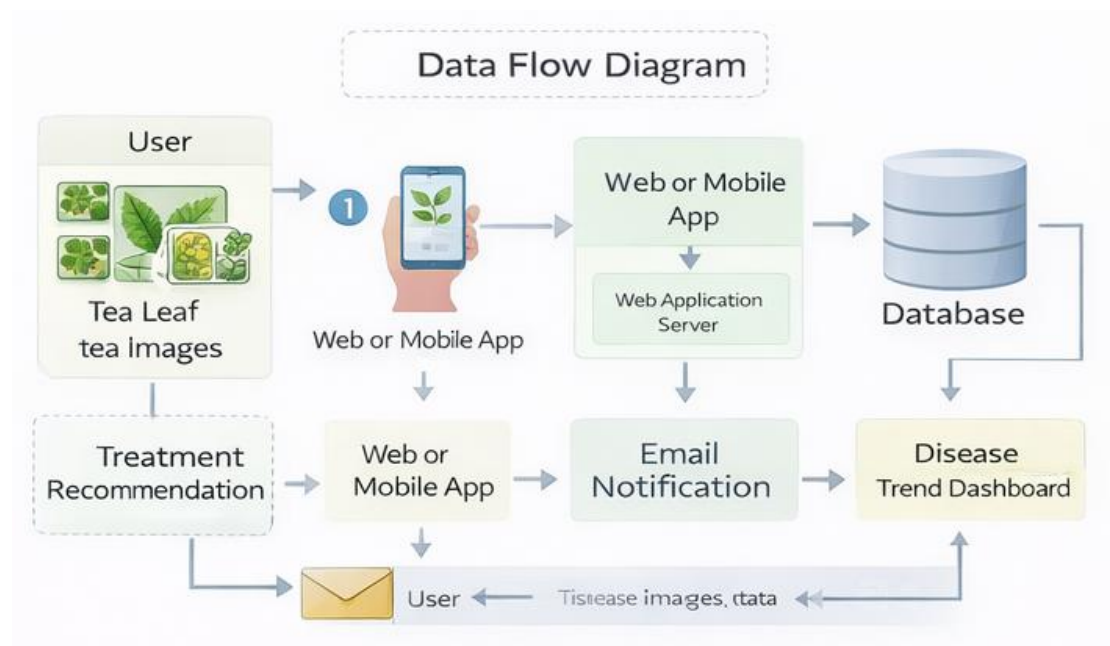


Figure 11.2.3– Data Flow Diagram

As shown in Figure 11.2.3, the data flow begins when the user submits tea leaf images through the web or mobile application. These images are routed to the Web Application Server, which processes them and stores all inputs and outputs in the central database. On the return path, the system delivers treatment recommendations back to the user interface and simultaneously dispatches email notifications to relevant stakeholders such as estate managers. A dedicated disease trend dashboard aggregates all stored results to provide visual analytics on outbreak frequency, seasonal patterns, and location-based disease hotspots.

Appendix Figure 11.2.4 illustrates the disease pattern analysis diagram, showing how historical detection data is processed to identify seasonal trends, location-based hotspots, and recurring outbreak patterns.

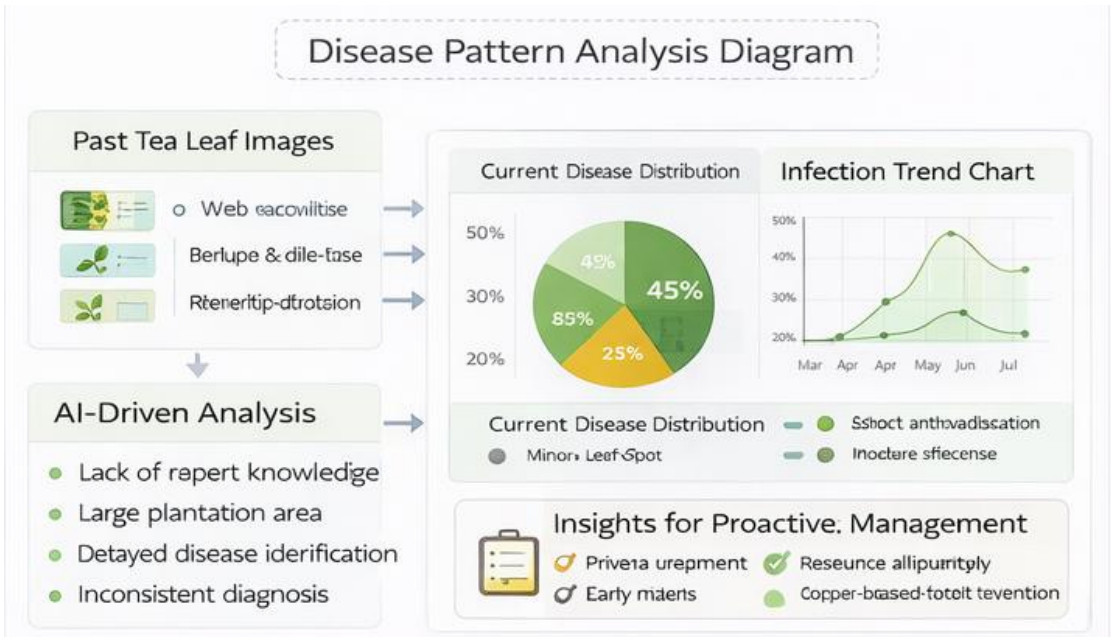


Figure 11.2.4– Disease Pattern Analysis Diagram

Figure 11.2.4 demonstrates how accumulated historical detection records are transformed into actionable insights through AI-driven analysis. Past tea leaf images and their associated detection results feed into an analysis engine that generates three key outputs: a current disease distribution chart showing the relative prevalence of each disease type across the estate, an infection trend line chart tracking how disease incidence has changed over consecutive months, and a proactive management insight panel that recommends specific interventions such as preventive treatments, resource reallocation, and early monitoring schedules based on the identified patterns

11.3 Preliminary UI mockups

Appendix Figure 11.3.1 presents the preliminary UI mockup for the proposed system, illustrating the main screens and user interface layout for the disease detection and reporting features.

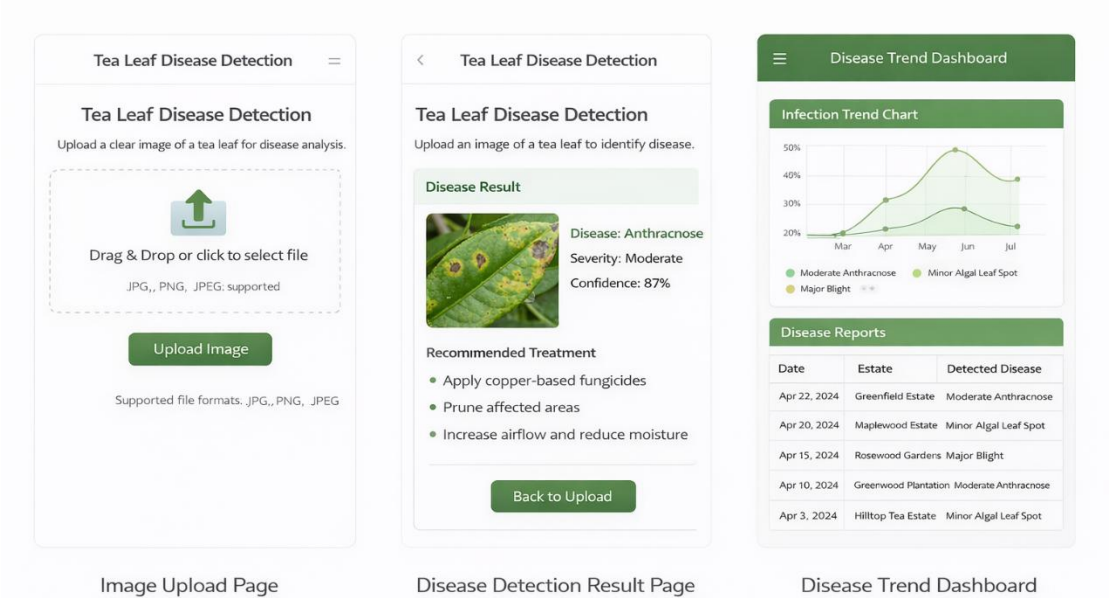


Figure 11.3.1 Preliminary UI mockups

Figure 11.3.1 shows three primary screens of the proposed application. The Image Upload Page on the left provides a simple drag-and-drop interface where users can select or drop a leaf image in JPG, PNG, or JPEG format before tapping the upload button. The Disease Detection Result Page in the center displays the analysis output immediately after processing, presenting the detected disease name, severity level, and confidence score alongside a thumbnail of the submitted leaf, followed by a list of recommended treatments such as applying copper-based fungicides, pruning affected areas, and increasing airflow. The Disease Trend Dashboard on the right gives estate managers a high-level overview of plantation health, showing an infection trend chart across months and a disease report table that lists recent detections by date, estate name, and identified disease type, enabling quick identification of outbreak patterns across multiple plantation locations

11.4 Risk analysis

Table 11.4 presents the risk analysis for the proposed research, identifying potential risks, their impact on the project, and the mitigation strategies planned to address each risk.

Table 11.4 Risk analysis

Risk ID	Risk Description	Impact	Mitigation Strategy
R1	Insufficient or unbalanced tea leaf disease dataset	May reduce CNN model accuracy	Collect additional images from multiple sources and use data augmentation techniques
R2	Poor image quality captured by users	Incorrect disease detection results	Apply image preprocessing such as resizing, noise reduction, and normalization
R3	Overfitting of CNN model during training	Model performs poorly on new data	Use regularization, dropout layers, and cross-validation
R4	Limited access to tea estates for field data collection	Difficulty obtaining real-world images	Use publicly available plant disease datasets and collaborate with local estates
R5	System performance limitations during image processing	Slow prediction response time	Optimize model architecture and use cloud-based GPU resources
R6	Data privacy concerns related to uploaded images	Ethical and legal concerns	Ensure anonymization and secure data storage

As presented in Table 11.4, the identified risks primarily relate to dataset quality, model performance, and data collection access. Each risk has been assigned a mitigation strategy to ensure the research can be completed successfully within scope.

11.5 Ethical documents

The proposed research titled **“Tea Leaf Disease Detection and Pattern Analysis using Convolutional Neural Networks (CNN)”** will follow ethical research standards throughout the data collection, analysis, and system development phases.

Informed Consent

Any tea leaf images or plantation-related information collected from tea estates will be obtained with the permission of estate management or supervisors. If interviews or surveys are conducted with estate staff, participants will be informed about the purpose of the research and how the collected information will be used before participation.

Data Privacy and Confidentiality

The research will not collect or store personal or sensitive information related to tea estate workers or staff. All collected data will be used strictly for academic research purposes. Any identifiable information related to tea estates will be anonymized to ensure confidentiality.

Responsible Use of Data

Datasets used for training and evaluation of the CNN model will be obtained either through authorized field data collection or publicly available datasets from research repositories. Proper citations and acknowledgements will be provided for all external data sources used in the research.

Data Security

All collected images and analysis results will be stored securely within the system database. Access to research data will be limited to the research team to ensure data integrity and protection from unauthorized access.

Ethical Research Compliance

This research will comply with the ethical research guidelines of the Sri Lanka Institute of Information Technology (SLIIT) and will ensure that the system supports sustainable agricultural practices without causing harm to individuals, organizations, or the environment.